



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Mid Term Examinations (Fall 2022) (SOLUTION)

Course Code with Title	CS-215T: Computer Organization & Assembly Language		Program	Bachelor of Computer Science BS(CS)
Instructor(s)	Mr. Shakir Karim / Dr. Abdul Hameed Pitafi		Semester	3 rd
Start date & Time	December 09, 2022 at 11:15 AM	End Time	12:45 PM	
Maximum Marks	30			

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

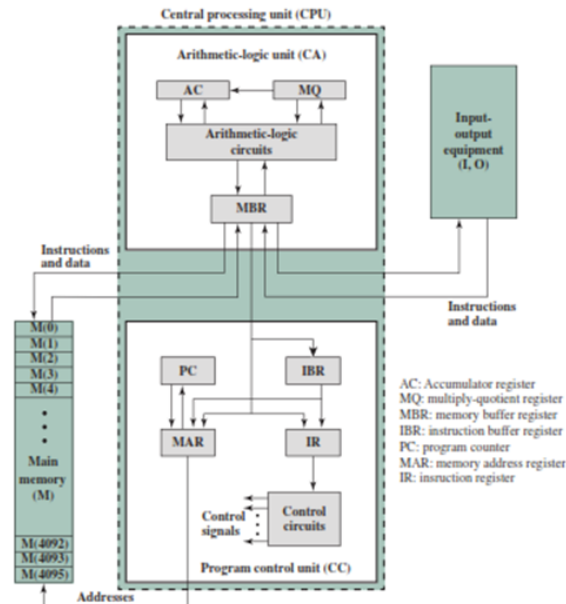
- Attempt All Questions.
- Use of calculator (simple/scientific/programmable) is not allowed.
- Mobile Phones are strictly prohibited in examination premises.

Q.1.

(03)

Illustrate the computer model proposed by von Neumann using block diagram.

Answer:





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Q.2. (05)

Show step-by-step standard process of signed multiplication to multiply **(-5)** and **(+6)**. Assume 4-bits registers to hold signed numbers.

Answer:

$$-5 = 1011$$

$$+6 = 0110$$

Iteration #	A ₃ A ₂ A ₁ A ₀	Q ₃ Q ₂ Q ₁ Q ₀	Q ₋₁	M ₃ M ₂ M ₁ M ₀	Remarks
Initializing	0 0 0 0	1 0 1 1	0	0 1 1 0	
Iteration #1	1 0 1 0 1 1 0 1	1 0 1 1 0 1 0 1	0 1	0 1 1 0	A=A-M SAR
Iteration #2	1 1 1 0	1 0 1 0	1	0 1 1 0	SAR
Iteration #3	0 1 0 0 0 0 1 0	1 0 1 0 0 1 0 1	1 0	0 1 1 0	A=A+M SAR
Iteration #4	1 1 0 0 1 1 1 0	0 1 0 1 0 0 1 0	0 1	0 1 1 0	A=A-M SAR

$$+6 \times -5 = -30 = 11100010_2 \text{ (2's complement of 30)}$$

OR

:

$$+6 = 0110$$

$$-5 = 1011$$

Iteration #	A ₃ A ₂ A ₁ A ₀	Q ₃ Q ₂ Q ₁ Q ₀	Q ₋₁	M ₃ M ₂ M ₁ M ₀	Remarks
Initializing	0 0 0 0	0 1 1 0	0	1 0 1 1	
Iteration #1	0 0 0 0	0 0 1 1	0	1 0 1 1	SAR
Iteration #2	0 1 0 1 0 0 1 0	0 0 1 1 1 0 0 1	0 1	1 0 1 1	A=A-M SAR
Iteration #3	0 0 0 1	0 1 0 0	1	1 0 1 1	SAR
Iteration #4	1 1 0 0 1 1 1 0	0 1 0 0 0 0 1 0	1 10	1 0 1 1	A=A+M SAR

$$-5 \times +6 = -30 = 11100010_2 \text{ (2's complement of 30)}$$

Q.3. (05)

Represent **121.0625₁₀** in **IEEE 754 single precision** format. Show all the necessary steps involved in the process.

Answer:

Since single precision is 32 bit format.

(01 Sign bit, 08 Biased Exponent bits, 23 Significand bits)

$$\text{Here, Bias} = 2^{8-1} - 1 = 2^7 - 1 = 128 - 1 = 127$$

$$121.0625_{10} = 1111001.0001_2 = 1.1110010001 \times 2^6$$

Sign bit = **0** since number is +ve

$$\text{Biased Exponent} = \text{True Exponent} + \text{Bias} = 6 + 127 = 133_{10} = 10000101_2$$

$$\text{Significand} = 11100100010000000000000_2$$

$$121.0625_{10} = 0100001011110010001000000000000_2$$

Q.4.

(03+03+03)

(i) **MOV AL, [BP+DI]** (ii) **PUSH DI** (iii) **MOV [1254H], BX**

- (i) Base-Indexed Addressing mode
Memory Address: $850F0 + (1000+560D) = 8B6FDh$
Content at $8B6FD = \text{unknown/garbage value}$; AL will be updated
($850F:660D$)
- (ii) Stack Addressing mode
Memory Address: $850F0 + (FFFD-2) = 950EBh$
Content at $950EBh = 0Dh$ and Content at $950ECh = 56h$
($850F:FFFB$) ($850F:FFFC$)
- (iii) Direct Addressing mode
Memory Address: $70AC0 + (1254) = 71D14h$
Content at $71D14h = 78h$ and Content at $71D15h = 56h$
($70AC:1254$) ($70AC:1255$)



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Q.5.

(02+02+02+02)

Refer to the table 1 and instruction set summary (for *MOV* and *ADD* instructions), write machine code for the given instructions.

(i) **ADD [BP+SI], AL**

(ii) **MOV [DI]+1234H, DL**

(iii) **MOV DS, 5434H**

(iv) **ADD [DI], BL**

Answer:

(i) **Memory <= Register**

000000dw mod reg r/m

d=0 since register is source

w=0 since register is 8 bits wide

mod=00 since memory without displacement is used

reg=000 since AL is used as register

r/m=010 since [BP+SI]

00000000 00 000 010 = 0002h

(ii) **Memory <= Register**

100010dw mod reg r/m

d=0 since register is source

w=0 since register is 8 bits wide

mod=10 since memory with 16bits displacement is used

reg=010 since DL is used as register

r/m=101 since [DI]+16bit

10001000 10 010 101 0011010000010010 = 88953412h

(iii) **ILLEGAL INSTRUCTION**

(iv) **Memory <= Register**

000000dw mod reg r/m

d=0 since register is source

w=0 since register is 8 bits wide

mod=00 since memory without displacement is used

reg=011 since BL is used as register

r/m=101 since [DI]

00000000 00 011 101 = 001Dh



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Table 1: R/M and MOD values

R/M \ MOD	00	01	10	11	
	0-bit displacement	8-bit displacement	16-bit displacement	w=0	w=1
000	[BX+SI]	[BX+SI]+D8	[BX+SI]+D16	AL	AX
001	[BX+DI]	[BX+DI]+D8	[BX+DI]+D16	CL	CX
010	[BP+SI]	[BP+SI]+D8	[BP+SI]+D16	DL	DX
011	[BP+DI]	[BP+DI]+D8	[BP+DI]+D16	BL	BX
100	[SI]	[SI]+D8	[SI]+D16	AH	SP
101	[DI]	[DI]+D8	[DI]+D16	CH	BP
110	DIRECT	[BP]+D8	[BP]+D16	DH	SI
111	[BX]	[BX]+D8	[BX]+D16	BH	DI

8088



8086/8088 Instruction Set Summary

Mnemonic and Description	Instruction Code			
DATA TRANSFER				
MOV = Move:	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0
Register/Memory to/from Register	1 0 0 0 1 0 d w	mod reg r/m		
Immediate to Register/Memory	1 1 0 0 0 1 1 w	mod 0 0 0 r/m	data	data if w = 1
Immediate to Register	1 0 1 1 w reg	data	data if w = 1	
Memory to Accumulator	1 0 1 0 0 0 0 w	addr-low	addr-high	
Accumulator to Memory	1 0 1 0 0 0 1 w	addr-low	addr-high	
Register/Memory to Segment Register	1 0 0 0 1 1 1 0	mod 0 reg r/m		
Segment Register to Register/Memory	1 0 0 0 1 1 0 0	mod 0 reg r/m		
ARITHMETIC				
ADD = Add:	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0
Reg./Memory with Register to Either	0 0 0 0 0 0 d w	mod reg r/m		
Immediate to Register/Memory	1 0 0 0 0 s w	mod 0 0 0 r/m	data	data if s:w = 01
Immediate to Accumulator	0 0 0 0 0 1 0 w	data	data if w = 1	